

Multirate DSP:

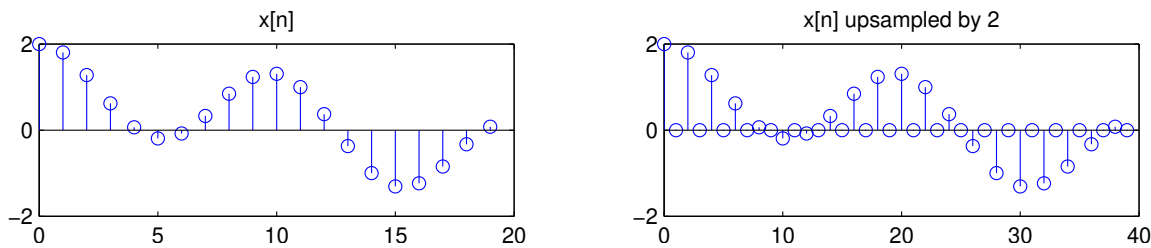
- Previously, we learned about several multirate operations:
 - downsampling
 - upsampling
 - decimation
 - interpolation
 - resampling
- We will now revisit and expand upon those operations via
 - z -domain derivations
 - filter design for interpolation/decimation
 - the Noble identities
 - polyphase interpolation, decimation, and resampling
 - allpass interpretation of ideal polyphase filters
 - multistage interpolation

Upsampling:

- Recall the upsampling operation

$$x[n] \longrightarrow \boxed{\uparrow U} \longrightarrow y[m]$$

- To upsample $x[n]$ by $U \in \mathbb{Z}^+$, we simply insert $U - 1$ zeros between each sample in $x[n]$.



- In the time domain, upsampling can be written as

$$y[m] = \begin{cases} x[\frac{m}{U}] & \text{if } \frac{m}{U} \in \mathbb{Z} \\ 0 & \text{else} \end{cases}$$

- In the DTFT domain, we previously derived that

$$Y(e^{j\omega}) = X(e^{j\omega U}) = X((e^{j\omega})^U)$$

- We can convert this to the z -domain via $e^{j\omega} \rightarrow z$:

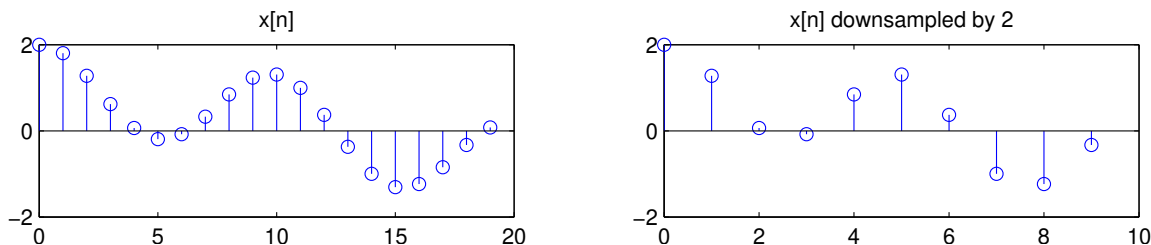
$$Y(z) = X(z^U)$$

Downsampling:

- Recall the downsampling operation

$$x[n] \longrightarrow \boxed{\downarrow D} \longrightarrow y[m]$$

- To downsample $x[n]$ by $D \in \mathbb{Z}^+$, we keep every D^{th} sample of $x[n]$ and discard the others.



- In the time domain, downsampling can be written as

$$y[m] = x[Dm]$$

- In the DTFT domain, we previously derived that

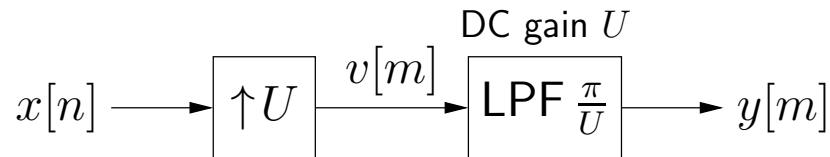
$$Y(e^{j\omega}) = \frac{1}{D} \sum_{k=0}^{D-1} X(e^{j\frac{\omega-2\pi k}{D}}) = \frac{1}{D} \sum_{k=0}^{D-1} X((e^{j\omega})^{\frac{1}{D}} e^{-j\frac{2\pi k}{D}})$$

- We can convert this to the z -domain via $e^{j\omega} \rightarrow z$:

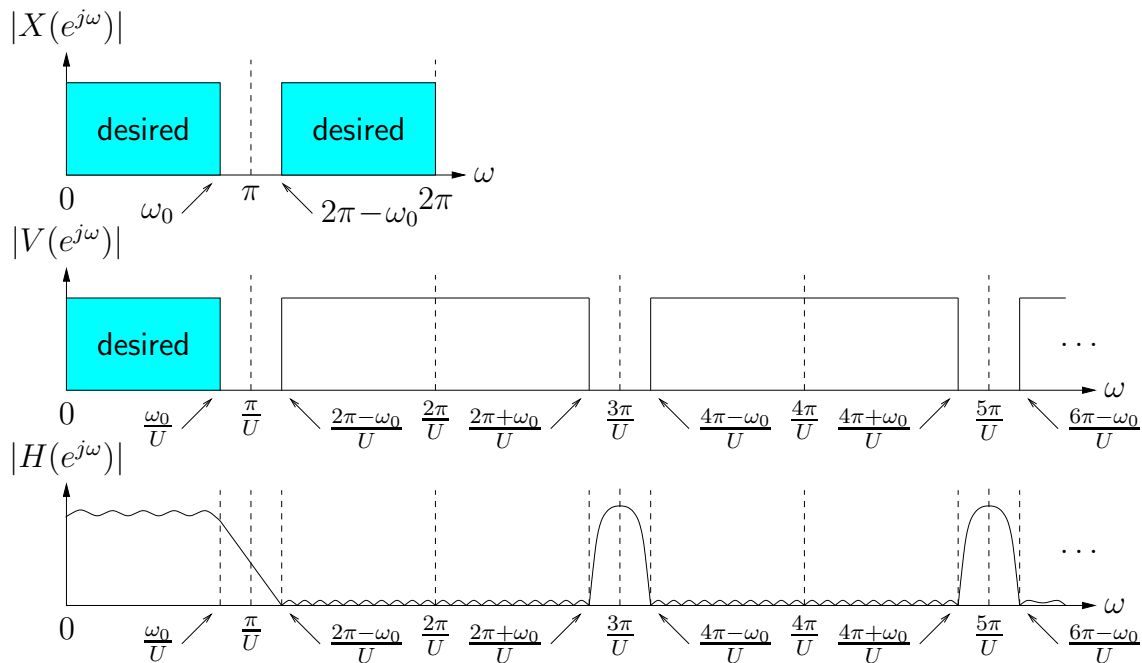
$$Y(z) = \frac{1}{D} \sum_{k=0}^{D-1} X(z^{\frac{1}{D}} e^{-j\frac{2\pi k}{D}})$$

Interpolation Filter Design:

- Recall interpolation, which increases the effective sampling rate by a factor $U \in \mathbb{Z}^+$ using



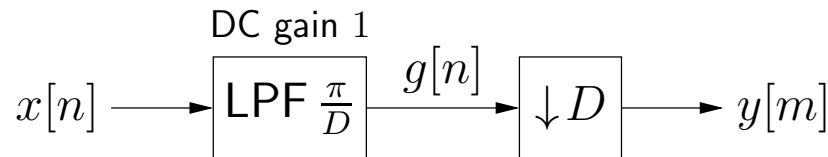
- The LPF removes unwanted spectral copies in $V(e^{j\omega})$. Let's take a closer look at the design of this LPF.
- Assuming real $x[n]$ bandlimited to ω_0 radians/sample, we can treat the dead zones in $V(e^{j\omega})$ as transition bands:



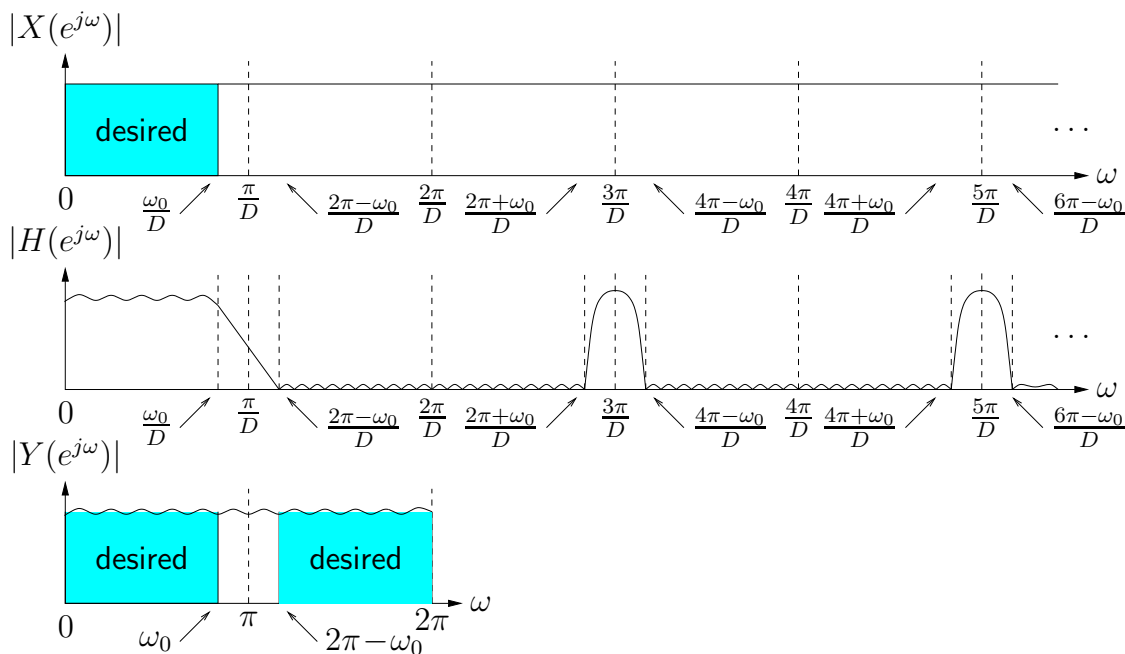
- By including these transition bands, we can design a filter with fewer coefficients and/or better ripple performance.

Decimation Filter Design:

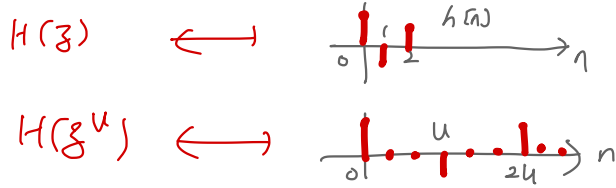
- Recall decimation, which decreases the effective sampling rate by factor $D \in \mathbb{Z}^+$ using



- The LPF is used to prevent aliasing when downsampling.
- Assuming we want to preserve frequencies $|\omega| \leq \frac{\omega_0}{D}$ in $X(e^{j\omega})$, we can place transition bands at frequencies that won't alias into frequencies $|\omega| \leq \omega_0$ in $Y(e^{j\omega})$:



- Note that the passband error in $Y(e^{j\omega})$ will approximately equal $\frac{1}{D}(\delta_p + (D-1)\delta_s)$, where $\frac{1}{D}$ is due to the D -downsampling, and $D-1$ is the number of stopbands in $G(e^{j\omega})$ that alias into $|\omega| \leq \omega_0$ in $Y(e^{j\omega})$.



The Noble Identities:

- The Noble identities tell us when we can swap the order of filtering and upsampling or downsampling.
- We will use them to derive efficient “polyphase” approaches to interpolation and decimation.
- The *Noble identity for upsampling* is the following:



Note that the impulse response of $H(z^U)$ is the U -upsampled version of $\{h[n]\}$.

- Proof: In the left diagram above, we have

$$Y(z) = H(z^U)V_1(z) = H(z^U)X(z^U)$$

while in the right diagram, we have

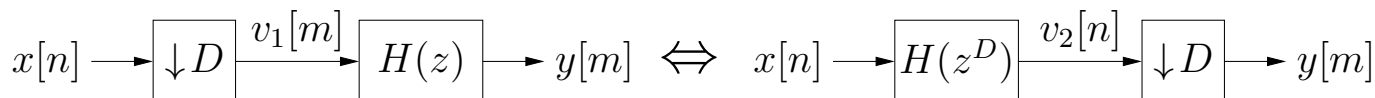
$$Y(z) = V_2(z^U) = H(z^U)X(z^U) \quad \text{since} \quad V_2(z) = H(z)X(z)$$

Thus, the left and right diagrams are equivalent.

- In words, this identity says “filtering-then-upsampling is equivalent to upsampling-then-filtering with an upsampled filter.”

The Noble Identities (cont.):

- The *Noble identity for downsampling* is the following:



Similar to before, the impulse response of $H(z^D)$ is the D -upsampled version of $\{h[n]\}$.

- Proof: In the left diagram above, we have

$$Y(z) = H(z)V_1(z) = H(z)\frac{1}{D}\sum_{k=0}^{D-1}X(e^{-j\frac{2\pi}{D}k}z^{\frac{1}{D}})$$

while in the right diagram, we have

$$\begin{aligned} Y(z) &= \frac{1}{D}\sum_{k=0}^{D-1}V_2(e^{-j\frac{2\pi}{D}k}z^{\frac{1}{D}}) \text{ where } V_2(z) = X(z)H(z^D) \\ &= \frac{1}{D}\sum_{k=0}^{D-1}X(e^{-j\frac{2\pi}{D}k}z^{\frac{1}{D}})H(e^{-j2\pi k}z) \\ &= H(z)\frac{1}{D}\sum_{k=0}^{D-1}X(e^{-j\frac{2\pi}{D}k}z^{\frac{1}{D}}) \end{aligned}$$

Thus, the left and right diagrams are equivalent.

- In words, this identity says “downsampling-then-filtering is equivalent to filtering-then-downsampling with an upsampled filter.”

Polyphase Interpolation:

- The standard interpolation procedure

$$x[n] \longrightarrow \boxed{\uparrow U} \longrightarrow \boxed{H(z)} \longrightarrow y[m]$$

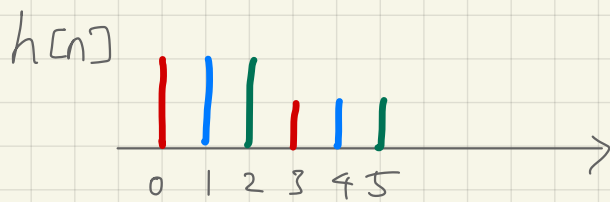
is computationally inefficient because the filter operates on a sequence of mostly zeros.

- Using the Noble identities, we can design an efficient interpolator.
- To begin, we decompose the filter $H(z)$ into its polyphase components $\{H_p(z)\}_{p=0}^{U-1}$:

$$\begin{aligned}
 H(z) &= \sum_{n \in \mathbb{Z}} h[n] z^{-n} \\
 &= \sum_{m \in \mathbb{Z}} \sum_{p=0}^{U-1} h[mU + p] z^{-(mU+p)} \quad \text{via } m \triangleq \lfloor \frac{n}{U} \rfloor, p \triangleq \langle n \rangle_U \\
 &= \sum_{p=0}^{U-1} \left(\sum_{m \in \mathbb{Z}} h_p[m] z^{-mU} \right) z^{-p} \quad \text{via } h_p[m] \triangleq h[mU + p] \\
 &= \sum_{p=0}^{U-1} H_p(z^U) z^{-p} \quad \text{p-th polyphase component}
 \end{aligned}$$

$h[n] \rightarrow h[n+p]$ shift to the left by p
 $\downarrow$
 $h[mU+p]$ down sample by factor U
 $(U=3)$
 $n = \begin{matrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \\ 9 & 10 & 11 \end{matrix} \begin{matrix} m \\ \downarrow \\ p \end{matrix} \rightarrow$

- The filter $\{h_p[m]\}_{m \in \mathbb{Z}}$ contains every U th sample of $\{h[n]\}_{n \in \mathbb{Z}}$ at offset p .

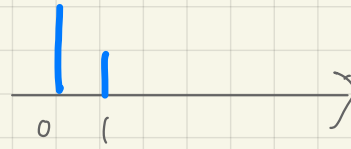


$$\begin{aligned}
 h[n] &\rightarrow \boxed{\downarrow 3} \rightarrow h_1[n] \\
 h[n] &\rightarrow \boxed{z} \rightarrow \boxed{\downarrow 3} \rightarrow h_2[n] \\
 h[n] &\rightarrow \boxed{z^2} \rightarrow \boxed{\downarrow 3} \rightarrow h_3[n]
 \end{aligned}$$

$$h_1[n] \leftrightarrow H_1(z)$$



$$h_2[n] \leftrightarrow H_2(z)$$



$$h_3[n] \leftrightarrow H_3(z)$$



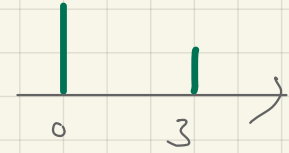
$$\tilde{h}_1[n] \leftrightarrow H_1(z^3)$$



$$\tilde{h}_2[n] \leftrightarrow H_2(z^3)$$



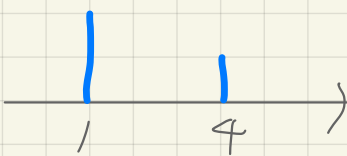
$$\tilde{h}_3[n] \leftrightarrow H_3(z^3)$$



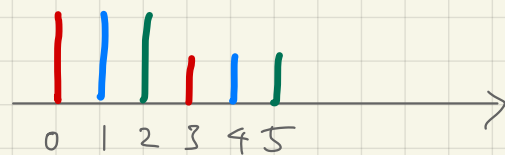
$$\tilde{h}_1[n] \leftrightarrow H_1(z^3)$$



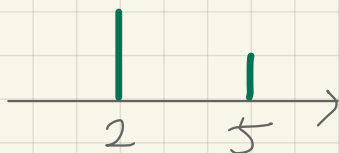
$$\hat{h}_2[n-1] \leftrightarrow z^{-1} H_2(z^3)$$



$$h[n] \leftrightarrow H(z)$$



$$\tilde{h}_3[n-2] \leftrightarrow z^{-2} H_3(z^3)$$



$$H(z) = H_1(z^3) + z^{-1} H_2(z^3) + z^{-2} H_3(z^3)$$